

DASHBALBAR, MongoliaWMO: **442560**Lat: **49.5508N**Lon: **114.4042E**Elev: **2313**StdP: **13.51**Time Zone: **8.00 (E08)**Period: **95-19**WBAN: **99999****Annual Heating, Humidification, and Ventilation Design Conditions**

Coldest Month	Heating DB		Humidification DP/MCDB and HR						Coldest Month WS/MCDB				MCWS/PCWD to 99.6% DB		WSF
			99.6%			99%			0.4%		1%				
	99.6%	99%	DP	HR	MCDB	DP	HR	MCDB	WS	MCDB	WS	MCDB	MCWS	PCWD	
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
1	-28.7	-24.1	-32.4	1.0	-27.4	-28.1	1.2	-23.3	27.2	-0.2	23.2	3.3	1.0	270	0.705

Annual Cooling, Dehumidification, and Enthalpy Design Conditions

Hottest Month	Hottest Month DB Range	Cooling DB/MCWB						Evaporation WB/MCDB						MCWS/PCWD to 0.4% DB	
		0.4%		1%		2%		0.4%		1%		2%			
		DB	MCWB	DB	MCWB	DB	MCWB	WB	MCDB	WB	MCDB	WB	MCDB	MCWS	PCWD
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
7	20.8	89.4	67.8	85.4	66.8	81.7	65.5	71.1	84.7	68.9	81.2	67.0	78.1	10.0	180

Dehumidification DP/MCDB and HR									Enthalpy/MCDB						Extreme Max WB
0.4%			1%			2%			0.4%		1%		2%		
DP	HR	MCDB	DP	HR	MCDB	DP	HR	MCDB	Enth	MCDB	Enth	MCDB	Enth	MCDB	
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	
66.5	106.5	78.2	64.7	99.9	75.1	63.1	94.1	73.0	36.7	84.6	34.6	81.5	33.0	78.2	79.3

Extreme Annual Design Conditions

Extreme Annual WS			Extreme Annual Temperature				n-Year Return Period Values of Extreme Temperature									
			Mean		Standard Deviation		n=5 years		n=10 years		n=20 years		n=50 years			
1%	2.5%	5%	Min	Max	Min	Max	Min	Max	Min	Max	Min	Max	Min	Max	DB	WB
(a)	(b)	(c)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)		
(4) 27.7	23.2	21.4	-33.9	95.7	5.7	4.8	-38.0	99.1	-41.3	101.9	-44.5	104.6	-48.7	108.1		
(5)			-34.3	74.0	5.6	2.9	-38.3	76.1	-41.6	77.8	-44.8	79.5	-48.9	81.6		

Monthly Climatic Design Conditions

		Annual	Jan	Feb	Mar	Apr	May	Jun	Jul	Aug	Sep	Oct	Nov	Dec		
		(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)		
(6)	Temperatures, Degree-Days and Degree-Hours	DBAvg	33.4	-5.5	2.5	18.6	37.3	51.4	63.6	68.7	64.4	51.9	34.3	13.2	-1.0	(6)
(7)		DBStd	27.46	9.58	10.20	11.26	9.12	8.03	7.11	5.26	6.40	8.04	9.18	11.64	9.69	(7)
(8)		HDD50	7749	1720	1330	973	395	79	3	0	1	71	491	1105	1581	(8)
(9)		HDD65	11795	2185	1750	1438	832	429	106	19	87	398	950	1555	2046	(9)
(10)		CDD50	1706	0	0	0	12	122	411	580	447	128	6	0	0	(10)
(11)		CDD65	279	0	0	0	0	7	64	133	69	6	0	0	0	(11)
(12)		CDH74	3490	0	0	0	13	192	992	1393	772	128	0	0	0	(12)
(13)		CDH80	1196	0	0	0	2	61	351	495	258	29	0	0	0	(13)
(14)	Wind	WSAvg	8.8	7.1	7.6	9.4	11.8	12.0	8.8	7.7	7.8	8.8	9.0	8.1	7.7	(14)
(15)	Precipitation	PrecAvg	10.6	0.1	0.1	0.1	0.2	0.7	1.8	3.3	2.6	1.1	0.4	0.1	0.2	(15)
(16)		PrecMax	19.8	0.4	0.3	0.4	1.0	2.9	7.1	9.8	6.5	3.0	1.5	0.7	0.3	(16)
(17)		PrecMin	4.6	0.0	0.0	0.0	0.0	0.0	0.2	0.7	0.6	0.0	0.0	0.0	0.0	(17)
(18)		PrecStd	3.8	0.1	0.1	0.1	0.2	0.6	1.4	2.2	1.4	0.7	0.4	0.2	0.1	(18)
(19)	Monthly Design Dry Bulb and Mean Coincident Wet Bulb Temperatures	0.4%	DB	23.0	37.0	54.9	74.7	87.3	94.8	96.0	92.0	84.1	68.1	47.3	27.4	(19)
(20)			MCWB	19.8	29.7	42.4	56.2	60.5	68.6	72.3	71.1	63.6	51.3	38.7	23.6	(20)
(21)		2%	DB	17.4	30.1	48.6	65.7	79.1	88.1	89.7	86.5	77.5	60.5	40.0	21.7	(21)
(22)			MCWB	15.2	25.9	37.9	49.7	58.5	66.4	68.4	68.1	60.8	47.6	33.5	19.1	(22)
(23)		5%	DB	12.9	24.7	42.1	59.5	73.2	83.8	85.2	82.0	72.5	54.4	34.7	17.4	(23)
(24)			MCWB	11.2	21.5	33.9	46.2	56.0	64.9	68.0	66.4	58.3	44.2	29.8	15.3	(24)
(25)		10%	DB	8.9	19.5	36.8	54.4	68.6	79.6	81.3	77.8	67.8	49.6	30.3	13.2	(25)
(26)			MCWB	7.4	17.0	30.3	43.3	53.6	63.1	66.7	64.5	55.6	41.0	26.6	11.5	(26)
(27)	Monthly Design Wet Bulb and Mean Coincident Dry Bulb Temperatures	0.4%	WB	19.8	30.4	43.6	57.6	65.4	73.2	75.2	73.1	65.8	52.6	40.2	24.2	(27)
(28)			MCDWB	22.2	35.8	55.2	72.4	83.0	91.2	90.8	88.8	81.0	65.6	46.7	27.2	(28)
(29)		2%	WB	15.3	25.8	38.4	51.2	60.3	68.9	71.8	70.0	62.4	47.9	34.6	19.3	(29)
(30)			MCDWB	17.4	29.9	48.0	64.3	75.4	83.7	84.5	83.3	74.3	59.4	40.6	21.6	(30)
(31)		5%	WB	11.2	21.5	34.3	47.0	57.1	66.5	69.6	67.7	59.6	44.8	31.1	15.4	(31)
(32)			MCDWB	12.7	24.7	42.0	57.9	71.2	79.9	81.5	78.7	69.4	53.8	35.6	17.4	(32)
(33)		10%	WB	7.4	17.1	30.4	43.6	54.3	64.3	67.8	65.8	56.7	41.8	27.7	11.5	(33)
(34)			MCDWB	8.8	19.4	36.5	53.8	67.2	76.9	78.3	75.2	65.8	49.0	31.6	13.0	(34)
(35)	Mean Daily Temperature Range	5% DB	MDBR	20.1	24.2	24.4	25.0	25.9	24.1	20.8	21.5	24.0	22.3	20.0	19.0	(35)
(36)			MCDBR	22.6	28.2	30.4	33.8	35.0	32.0	28.4	29.3	32.5	30.0	24.5	20.5	(36)
(37)			MCWBR	20.8	24.5	21.8	22.0	19.7	15.5	12.9	14.3	18.1	19.4	17.8	18.5	(37)
(38)		5% WB	MCDBR	22.5	28.2	29.7	31.7	31.8	28.4	24.4	25.0	28.3	28.5	23.7	20.6	(38)
(39)			MCWBR	20.7	24.7	21.7	21.8	19.5	15.6	12.7	13.2	16.3	19.0	18.4	18.6	(39)
(40)	Clear-Sky Solar Irradiance	taub		0.254	0.280	0.320	0.396	0.476	0.457	0.423	0.386	0.337	0.305	0.276	0.261	(40)
(41)		taud		2.274	2.240	2.221	2.084	1.898	2.077	2.250	2.329	2.405	2.409	2.327	2.225	(41)
(42)		Ebn at Noon		255	274	280	267	249	256	264	269	273	263	243	231	(42)
(43)		Edh at Noon		22	30	36	47	59	49	41	36	30	25	21	20	(43)
(44)	All-Sky Solar Radiation	RadAvg		553	939	1408	1739	1969	2054	1896	1704	1370	928	564	416	(44)
(45)		RadStd		29	42	82	104	102	146	118	110	77	41	30	35	(45)